

AI-CRV COMPOSITE RISK VERIFICATION

Digital Platform for Operational Risk Scoring & Intelligence

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- CONFIDENTIAL -

THE PROBLEM ?

CURRENT RISK SYSTEMS: MANUAL / STATIC / DISCONNECTED

STANDARD RISK MATRIX			HAZARD EFFECT / CONSEQUENCE								
Loss Type	1 Insignificant	2 Minor	3 Moderate	4 Major	5 Catastrophic						
Timeline	No impact on overall project timeline	May result in overall project timeline overrun of less than 5%	May result in overall project timeline overrun of between 5% and less than 20%	May result in overall project timeline overrun of between 20% and less than 50%	May result in overall project timeline overrun of 50% or more						
Budget	No impact on the budget of the project	May result in overall project budget overrun of less than 5%	May result in overall project budget overrun of between 5% and less than 20%	May result in overall project budget overrun of between 20% and less than 50%	May result in overall project budget overrun of 50% or more						
Investment Return – NPV loss	Less than RM	RM to less than RM50k	RM50k to less than RM100k	RM100k to RM500k	RM500k or more						
Quality	No impact on quality	Minor quality issues that can be addressed in a short timeframe with minimal interventions	Some quality issues that requires immediate management action	Significant quality issues that requires senior project management intervention	Significant quality issues that requires senior project management intervention with significant resource and cost implications for rework						
Safety / Health	First aid cases / Exposure to minor health risk	Medical treatment case / Exposure to major health risk	Lost time injury / Recoverable impact on health	Single fatality or loss of quality of life / Irreversible impact on health	Multiple fatalities / Impact on health ultimately fatal						
Environment	Minimal environmental harm - L1 incident	Minor environmental harm - L2 incident remediable short term	Serious environmental harm - L3 incident remediable within 12 months	Major environmental harm - L4 incident remediable over 12 months	Extreme environmental harm - L5 incident irreversible						
Legal & Regulatory	No legal impact	Minor legal concerns with minimal impact	Some legal concerns with manageable level of impact	Serious legal concerns with significant impact on operations	Legal non-compliance with risk of shutdown of operations with significant cost implications for rework						
Reputation / Social / Community	Slight impact - public awareness may exist but no public concern	Limited impact - local public concern	Considerable impact - regional public concern	National impact - national public concern	International impact - international public attention						
LIKELIHOOD			RISK RATING								
5 Almost Certain	The unwanted event has occurred frequently, has a 90% and higher probability of reoccurring	11 Medium	16 Significant	20 Significant	23 High	26 High					
4 Likely	The unwanted event has a probability of between 60% and less than 90% of occurring	7 Medium	12 Medium	17 Significant	21 High	24 High					
3 Possible	The unwanted event has a probability of between 30% and less than 60% of occurring	4 Low	8 Medium	13 Significant	18 Significant	22 High					
2 Unlikely	The unwanted event has a probability of between 1% and less than 30% of occurring	2 Low	5 Low	9 Medium	14 Significant	19 Significant					
1 Rare	The unwanted event has never occurred, has a probability of less than 1% of occurring	1 Low	3 Low	6 Medium	10 Medium	15 Significant					
Risk Rating		Risk Level		Guidelines for Risk Matrix							
21 to 25	(4) – High	Eliminate, avoid, implement specific action plans/procedures to manage & monitor									
13 to 20	(3) – Medium	Proactively manage									
6 to 12	(2) – Low	Actively manage									
1 to 5	(1) – Low	Monitor & manage as appropriate									
No.	Task / Activity	Hazard	Risk	L	C	RR	Controls and further action to reduce the risk	L	C	RR	Responsible persons
1	Delivering equipment to site and offloading rods and other	Man / machine interface	Injuries Fatalities	2	4	14	• Driver must be escorted on site	1	4	10	Foreman Supervisor Driver

Department / Area	ZAULIYAH GAS PLANT (ZGP)	Severity	A	B	C	D	E	Likelihood	Approver
Job Type / Equipment / System	Operation	0						Low	DTL or Delegate
Process	Operation	1						Medium	Operations Manager
Task	As per below list	2						High	Oil, Gas or Engineering Director
Carried Out By -	ZGP Operation team	3						Serious	
Risk Assessment Tracking No.		4							
		5							

Based on HAZCON guidewords / SIMOPS Workshop:

No.	Activity / Task	Type of Hazard	Effect of the Hazard	Groups Affected	Initial Risk (Highest Rating)	Control measures required to reduce Initial Risk	Residual / Tolerable Risk (Severity / Likelihood)
1	Suspending the alarm of Condensate stabilizer column water level (84LZA705 C8470 COLUMN WTR LVL LL LTRP) in DCS due to no level or no water accumulation in the column bottom.	1. Overlooking alarm.	• No alarm in DCS.	ZGP Operator and working crew	Low (0A)	1. 84LZA705 will open 84UZV714 and 84LSA704 will control the water level if the column receives water. 2. Monitoring & logging the level once in a day.	Low (0A)
2	Suspending the alarm of Condensate Reboiler condensate outlet temperature (84TZAT14 - E8471 EXCHAL TEMP LL LTRP) in DCS due to low condensate production from the wells and higher handling capacity of Condensate Stabilizer Unit.	1. Overlooking alarm.	• No alarm in DCS. • CSU PSD if the low temperature alarm exists for 24hrs continuously.	ZGP Operator and working crew	Medium (1B)	1. Monitoring & logging the temperature of 84TZAT14 twice per shift. 2. Increasing the 84TZAT14 temperature gradually by 1°C if the TZAT14 in alarm for more than 12hrs.	Low (0A)
3	Suspending the alarm of Glycol/Condensate Heater outlet temperature (84TZAS13 E8432 COND TEMP LL) in DCS due to higher associated condensate (production) from EDP / Interlink's oil separated gas and condensing at chiller, and over handling capacity of Glycol/Condensate Heater.	1. Overlooking alarm.	• No alarm in DCS. • E-8432 trip if the low temperature alarm exists for 12hrs continuously.	ZGP Operator and working crew	Medium (1B)	1. Monitoring & logging the temperature of 84TZAS13 twice per shift. 2. Increasing the 84TZAS13 temperature gradually by 1°C if the TZAS13 in alarm for more than 06hrs.	Low (0A)
4	Suspending the below alarms of UPS in DCS for the time being due non-	1. Overlooking alarm.	• No alarm in DCS.	ZGP Operator and working crew	Low (0A)	1. Monitoring & logging for new UPS alarm once in shift.	Low (0A)

- 1 -

Revision 4.0



Manual Handling Assessment

Project/Facility: _____

Contact No: _____

Scope: _____

Assessed by: _____

Assessment Date: _____

Project Engineer: _____

Reference No: _____

Next Assessment Date: _____

Site Supervisor: _____

PART 1- ASSESSMENT (Answer the following questions)

	Yes	No
1. Does the operation involve a significant risk of injury? (complete Section B)		
2. If No the assessment need go no further.		
3. If Yes, can the operation be avoided, mechanized or level of risk reduced?		
4. If Yes, record steps in Sect. C & D and review		
5. Has the risk of injury been eliminated or reduced to an acceptable level?		
6. If Yes, the assessment is complete. If No, review activities to eliminate significant risks relating to Manual Handling Operations.		

<https://www.hsedocuments.com/>



PROBLEM OF STANDARD RISK SCORING

1. MULTIPLE **DISCONNECTED** SAFETY SYSTEMS
2. HUMAN OPERATORS ACT AS INTEGRATION LAYER
3. **COGNITIVE OVERLOAD** & ALARM **FATIGUE**
4. EQUAL WEIGHTING **IGNORES** DATA QUALITY
5. **STATIC** DECISION CYCLES
6. CRITICAL SIGNALS **DILUTED**
7. **No** TEMPORAL **FILTERING**
8. **No** EMERGENCY **OVERRIDE**

No COMPOSITE (Unified) VIEW

AI-CRV COMPOSITE RISK VERIFICATION

- REAL-TIME OPERATIONAL RISK INTELLIGENCE & SCORING FOR OIL & GAS, ENERGY INDUSTRY
- SINGLE RISK SCORE + AUTOMATED SAFETY OVERSIGHT

WHY AI-CRV SYSTEM EXISTS

1. CURRENTLY, YOUR OPERATIONS ARE MONITORED BY **MULTIPLE DISCONNECTED SYSTEMS**
2. HUMANS CANNOT PROCESS ALL SIGNALS IN REAL TIME
3. CRITICAL RISKS CAN BE MISSED OR DELAYED
4. NEED ONE TRUSTED, REAL-TIME SAFETY VIEW

*CORE AI-CRV LOGIC VALIDATED THROUGH 1,000+ STRESS-TEST SIMULATIONS.
CURRENTLY MOVING FROM SIMULATION-STAGE TO EDGE-HARDWARE INTEGRATION*

US PATENT PENDING



UNITED STATES
PATENT AND TRADEMARK OFFICE

Page 1 of 2

P.O. Box 1450
Alexandria, VA 22313 - 1450
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APPLICATION #

64/054,429

RECEIPT DATE / TIME

05/01/2026 06:24:30 AM Z ET

ATTORNEY DOCKET #

SITESHIELD-01-PROV

Title of Invention

AI-CRV: Artificial Intelligence Composite Risk Verification - A Domain Agnostic Method and System for Real-Time Multi-Criteria Industrial Safety Risk Assessment Using Confidence-Weighted Sensor Fusion with Asymmetric Override Logic

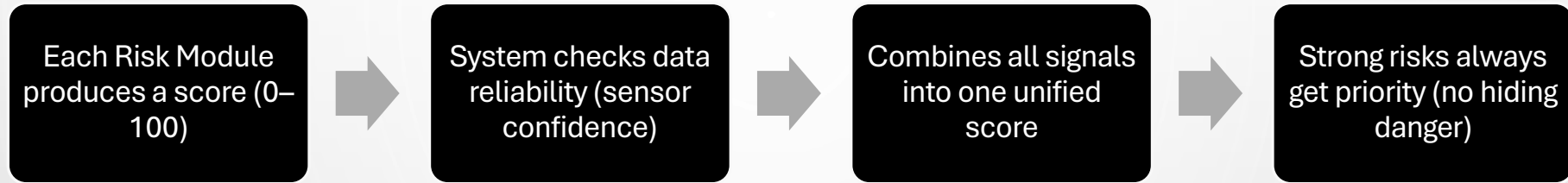
Application Information

APPLICATION TYPE

Utility - Provisional Application under
35 USC 111(b)



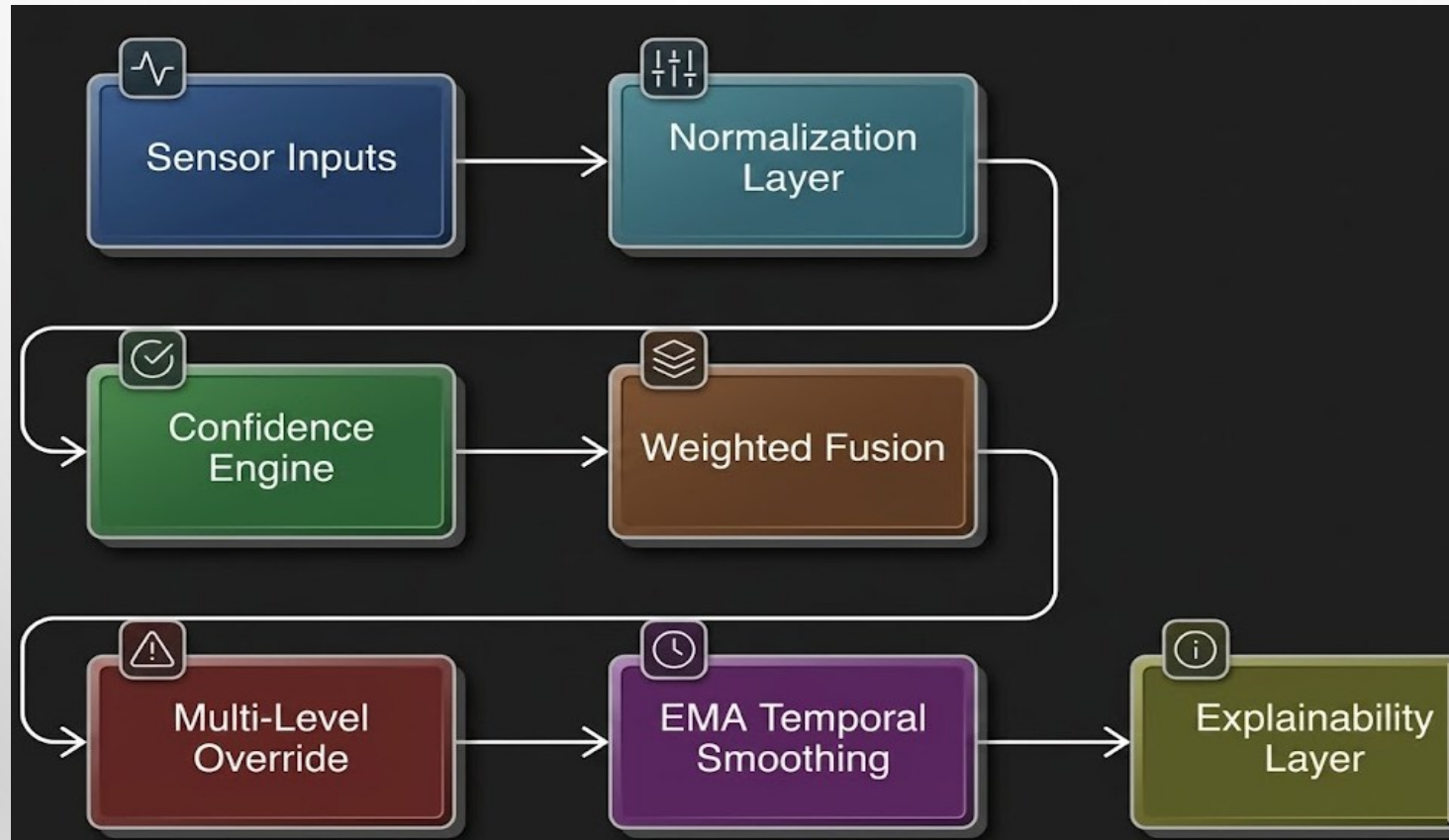
HOW IT WORKS



RISK INTELLIGENCE FEATURES

- **DETECTS SUDDEN CRITICAL EVENTS INSTANTLY**
- **FILTERS UNRELIABLE OR OUTDATED DATA**
- **PREVENTS ALARM NOISE AND FALSE ALERTS**
- **HIGHLIGHTS TOP DRIVERS OF RISK**
- **PROVIDES CONFIDENCE INDICATOR**

HOW IT WORKS



HOW IT WORKS – The Role of Ai

AI INTELLIGENCE LAYER (AI COMPONENTS)

AI models generate risk scores (0-100) and confidence for each risk domain



A.1 Tabular Models: XGBoost / LightGBM

- Predict likelihood of drone threat impacting installation
- **Inputs:** Historical incidents, platform profile, weather, operational context (crew activity, alerts)
- **Output:** Risk score (0-100), zone ranking, action suggestions
- **Example:** "Site Alpha scores 68/100 for drone intrusion risk in next 6 hours."



A.2 Time-Series Anomaly Detection: LSTM / GRU / Isolation Forest

- Detect anomalous radar/RF/sensor signals & drone behavior
- **Inputs:** Radar & RF streams, historical baselines, PTZ & thermal readings
- **Output:** Anomaly alerts, early warnings, confidence score



A.3 Computer Vision: YOLO-tiny

- Detect & classify drones and safety violations in real-time
- **Inputs:** PTZ, thermal/IR, in-cab, CCTV video feeds
- **Output:** Bounding boxes, class (drone type / PPE violation / zone intrusion), severity (L/M/H), confidence
- Edge inference with sub-second latency



A.4 Geospatial Probabilistic Models

Model Type	Description	Use in AI-CRV
Bayesian Networks	Graph-based probabilistic models (conditional dependencies)	Estimate attack likelihood given environment & operations
Monte Carlo Simulations	Randomized simulations of thousands of approach paths	Identify high-risk corridors & optimal sensor placement
Gaussian Processes / Kriging	Spatial interpolation of risk values across coordinates	Produce smooth continuous risk heatmaps
Poisson Point Processes	Model spatial occurrences of rare events	Estimate expected frequency of drone approaches per sector

WHAT OPERATORS SEE

One Unified Composite Risk Score (0–100)

Clear Alert Level (Normal → Emergency)

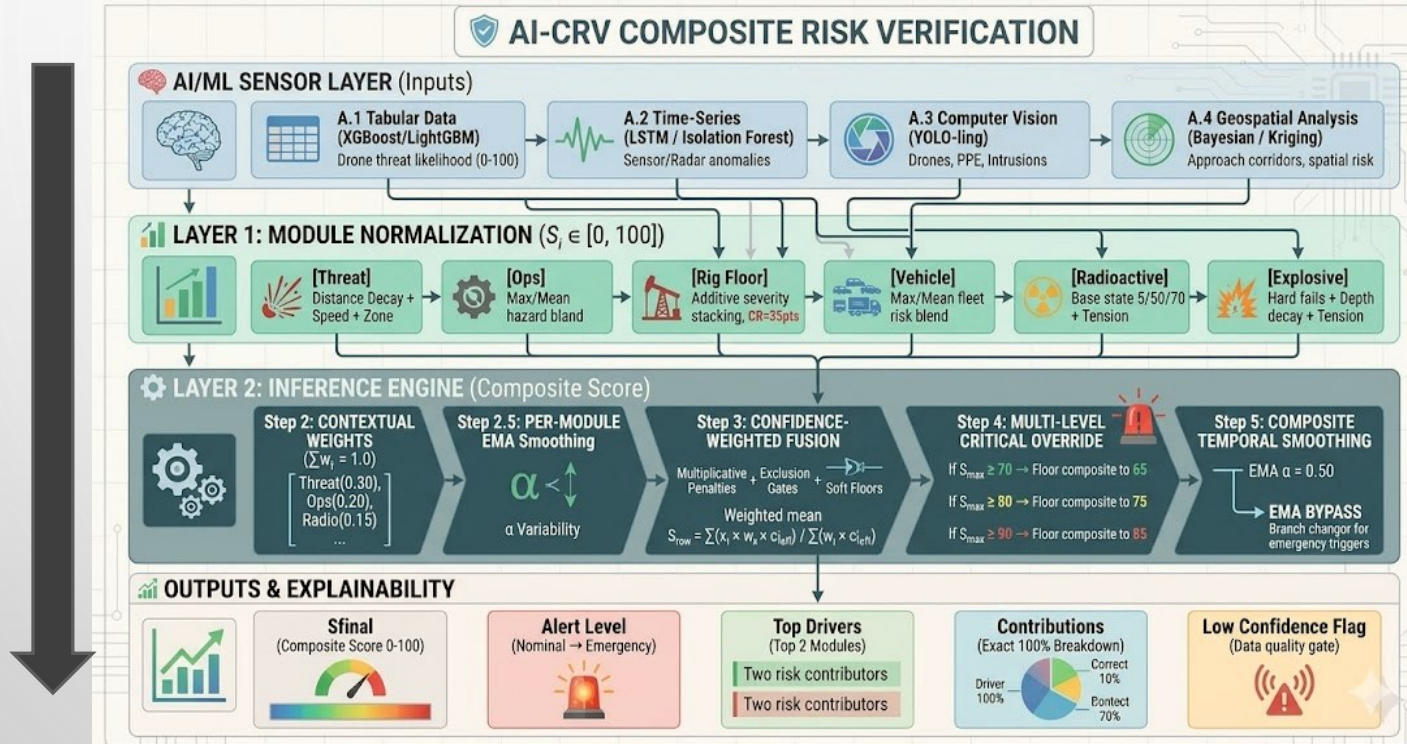
Instant immediate vs Smoothed Risk View

Explanation: what is driving risk



→ ***Faster, clearer operational decisions***

SYSTEM ARCHITECTURE DIAGRAM

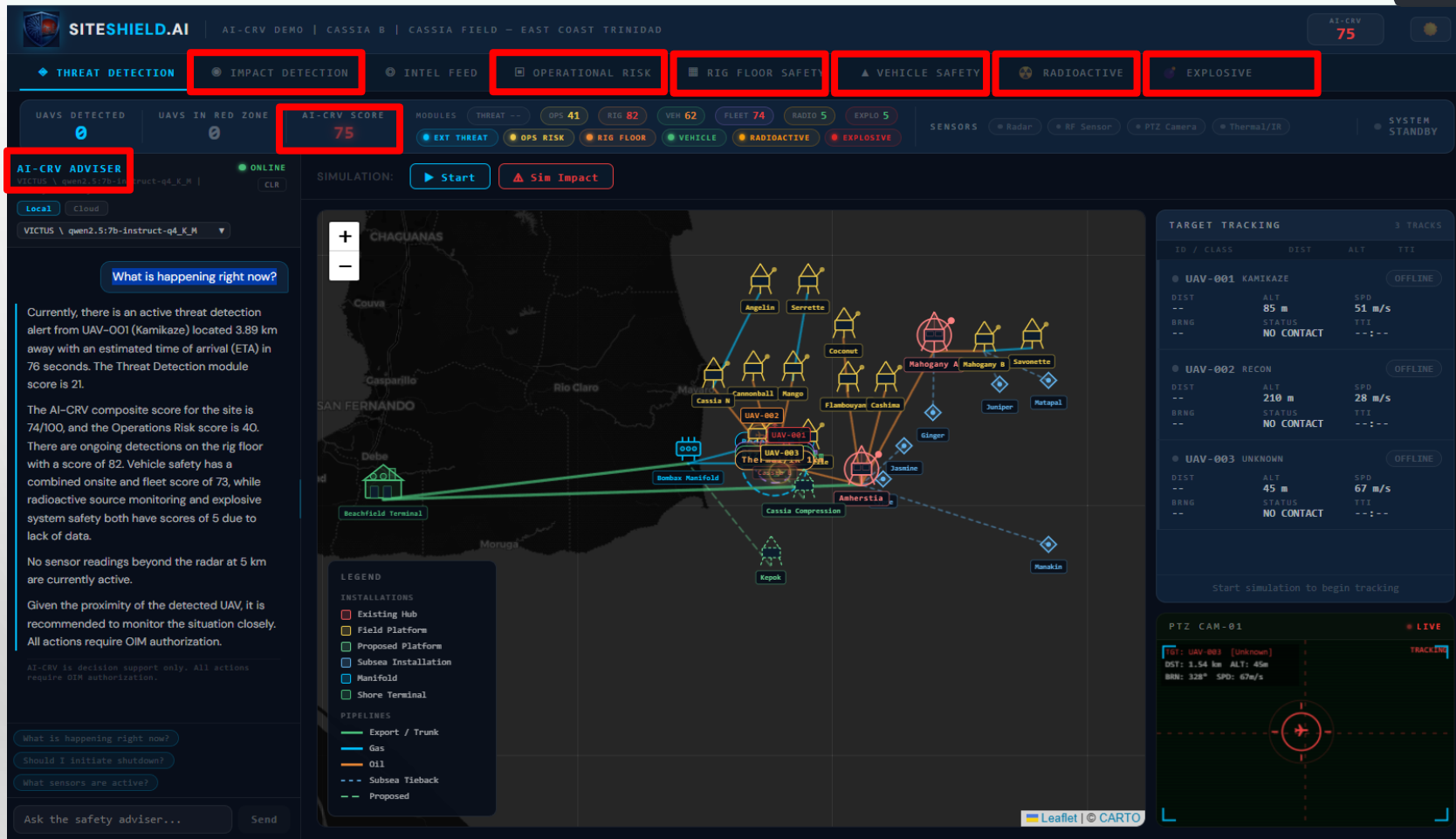


Unified Composite Risk Score (0–100)



APPLICATION PREVIEW (PRINT SCREENS)

MULTIPLE RISK MODULES (NATIVE + ADD-ON/ PLUG-IN) → COMPOSITE (Unified) Risk Score



Ai
Assistant
Side-panel



UAVS DETECTED 0

UAVS IN RED ZONE 0

AI-CRV SCORE 75

MODULES

THREAT --

OPS 42

RIG 82

VEH 62

FLEET 74

RADIO 5

EXPLO 5

SENSORS

Radar

RF Sensor

PTZ Camera

Thermal/IR

SYSTEM STANDBY

AI-CRV ADVISER

ONLINE

CLEAR

Local Cloud

VICTUS \ qwen2.5:7b-instruct-q4_K_M

Safety adviser ready.
Ask for a situation briefing
or use the quick prompts below.

- What is happening now?
- Should I initiate shutdown?
- What sensors are active?

OPERATION REGISTRY

1 operation on record

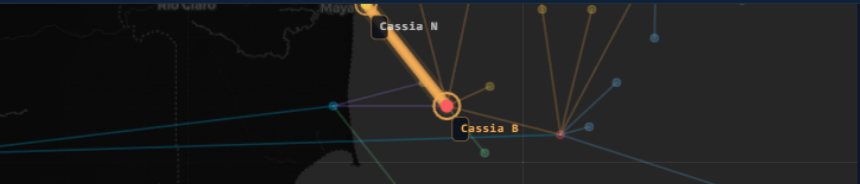
Pipeline Pigging — Segment 4B 42

Pigging ACTIVE

Phase: Pig in Run

Start: 06:30 | Est: 8 hours

OPS-2026-047



Pipeline Pigging — Segment 4B

Pigging Cassia N to Cassia B (Oil)

Field Supervisor — J. Al Rashidi

Start: 06:30 | Est. Duration: 8 hours

42 OPS RISK



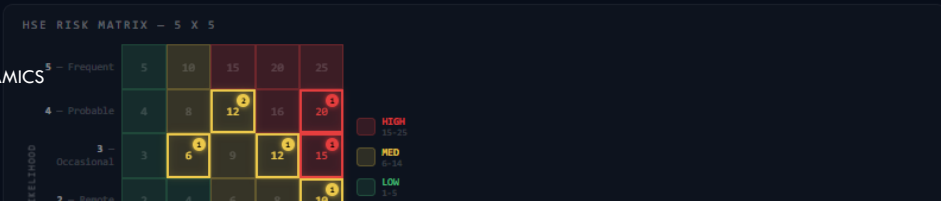
RISK RANKING SUMMARY

0 LOW

5 MEDIUM

2 HIGH

Active HIGH hazards present — engineering controls and full PPE mandatory before proceeding.



OPERATIONAL ALERTS

3 log entries

INFO 06:31

Pipeline Pigging OPS-2026-047 commenced — Pig in Run. BIRD detector active at all checkpoints

MEDIUM 07:42

H2S reading 12 ppm at receiver platform — below alarm threshold. Monitor continuously

MEDIUM 08:15

Flow velocity within acceptable range — pig stall risk LOW. Checkpoint 3 of 5 passed

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THREAT DETECTION

IMPACT DETECTION

INTEL FEED

OPERATIONAL RISK

RIG FLOOR SAFETY

VEHICLE SAFETY

RADIOACTIVE

EXPLOSIVE

UAVS DETECTED

0

UAVS IN RED ZONE

0

AI-CRV SCORE

85

MODULES

THREAT --

OPS 42

RIG 92

VEH 62

FLEET 74

RADIO 5

EXPLO 5

EXT THREAT

OPS RISK

RIG FLOOR

VEHICLE

RADIOACTIVE

EXPLOSIVE

SENSORS

Radar

RF Sensor

PTZ Camera

Thermal/IR

SYSTEM
STANDBY

AI-CRV ADVISER

ONLINE

VICTUS \ llava:7b | safety
advisory

CLR

Local

Cloud

VICTUS \ llava:7b

Safety adviser ready.

Ask for a situation briefing
or use the quick prompts below.

What is happening right now?

Should I initiate shutdown?

What sensors are active?

Ask the safety advise

Send

ZONE

Rig Floor 99

Monkey Board 91

Pipe Rack / Catwalk 45



PPE Violation Detection - Rig Floor

Ask AI



Rig Floor — CV Risk Assessment

Cameras: CAM-RF-01 | CAM-RF-02

Red Zones: V-Door, Rotary Table, Tong Swing Area

Model: YOLO-Tiny-v4 | Edge Inference

99

ZONE RISK

HIGH RISK

ACTIVE DETECTIONS

5 ACTIVE

CRITICAL FIRE / SMOKE

ACTIVE

Thermal anomaly detected near drill floor equipment

CAM-RF-01 | 14:54:03 | CONF: 81%

ACKNOWLEDGE

HIGH PPE VIOLATION

ACTIVE

Worker without safety vest detected near V-door

CAM-RF-02 | 14:53:43 | CONF: 83%

ACKNOWLEDGE

HIGH RED ZONE VIOLATION

ACTIVE

Personnel entered tong swing radius without clearance

CAM-RF-02 | 14:53:23 | CONF: 82%

ACKNOWLEDGE

HIGH RED ZONE VIOLATION

ACTIVE

Person detected in rotary table exclusion zone

CAM-RF-01 | 14:52:07 | CONF: 94%

ACKNOWLEDGE

HIGH PPE VIOLATION

ACTIVE

Worker missing hard hat near V-door area

CAM-RF-02 | 14:53:45 | CONF: 89%

ACKNOWLEDGE

CRITICAL FIRE / SMOKE

RESOLVED

Thermal anomaly detected near drill floor equipment

ZONE STATISTICS

5

ACTIVE

0

ACKNOWLEDGED

13

RESOLVED TODAY

87%

PPE COMPLIANCE

3

RED ZONE INC.

SITESHIELD.AI

AI-CRV DEMO | CASSIA B | CASSIA FIELD - EAST COAST TRINIDAD

AI-CRV85

THREAT DETECTION

IMPACT DETECTION

INTEL FEED

OPERATIONAL RISK

RIG FLOOR SAFETY

VEHICLE SAFETY

RADIOACTIVE

EXPLOSIVE

UAVS DETECTED0

UAVS IN RED ZONE0

AI-CRV SCORE85

MODULES

THREAT

OPS42

RIG99

VEH57

FLEET73

RADIO5

EXPLO5

SENSORS

Radar

RF Sensor

PIZ Camera

Thermal/IR

SYSTEMSTANDBY

AI-CRV ADVISER

ONLINE

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LocalCloud

VICTUS \ llava:7b

Safety adviser ready.

Ask for a situation briefing or use the quick prompts below.

VIEW MODE

ON-SITE

OFF-SITE

VEHICLE REGISTRY

7 vehicles

VH-001

Personnel Carrier

MOVING

DMS Feed 01

Gate 3 Monitoring Zone

35 km/h | Shift: 3.2h

VH-002

Supply Vehicle

MOVING

1 alert

DMS Feed 02

Perimeter Road East

28 km/h | Shift: 5.1h

VH-003

Company Personal Car

MOVING

1 alert

DMS Feed 03

Main Access Route

45 km/h | Shift: 6.8h

VH-004

Tanker Truck

MOVING

2 alerts

Driver A

Refinery Loading Bay 3

42 km/h | Shift: 6.2h

VH-005

Forklift

MOVING

1 alert

Driver B

Warehouse Zone C

8 km/h | Shift: 4.5h

Camera

Attentive

Awake

Seatbelt

Phone

Smoking

FPS: 30

Person ID: Tal

AOI: 5

Eye Mode: FIXATION (18)

Real Driver: YES

88.7

80.7

LIVE PARAMETERS

SPEED

35 km/h

SHIFT HOURS

3.2h

LAST BREAK

65 min

STATUS

Moving

ALERTS

0

DEPLOYMENT

ONSITE

RISK FACTOR BREAKDOWN

AFDB

Driver Fatigue

HIGH82

AFDB

Distraction — Phone Usage

HIGH74

Behavioral

Speed Compliance

LOW35

ALERT LOG

4 active | 5 total

FATIGUE ALERT

DMS Feed 03 - yawning detected. Driver on shift 6.8 hours.

VH-003

14:45:02

DISTRACTION

DMS Feed 02 - phone usage detected while vehicle in motion.

VH-002

14:44:38

FATIGUE ALERT

Driver A fatigue detected. Eyes closing. 6.2 hours on shift.

VH-004

14:44:12

DISTRACTION

Driver A - phone usage detected while moving.

VH-004

14:43:45

ZONE VIOLATION

Forklift entered pedestrian priority zone without clearance.

VH-005

14:38:28

FLEET ACTIONS

Contact Driver (VH-001)

Dispatch Supervisor

Halt Vehicle (Emergency)



UAVS DETECTED0

UAVS IN RED ZONE0

AI-CRV SCORE85

MODULES

THREAT --

OPS 42

RIG 99

VEH 55

FLEET 72

RADIO 48

EXPLO 5

SENSORS

Radar

RF Sensor

PTZ Camera

Thermal/IR

SYSTEM STANDBY

AI-CRV ADVISER

ONLINE

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CLR

Local

Cloud

VICTUS \ llava:7b

Safety adviser ready.
Ask for a situation briefing
or use the quick prompts below.

INCIDENT DEMO | UK OFFSHORE 2012 - TRAINING MODE

UK Offshore — Source Lost on Drill Floor

Worker failed to transfer a **50 GBq Cs-137** source from the transport container to the logging tool. The source lay unnoticed on the drill floor for **4–5 hours** while routine operations continued.

14WORKERS EXPOSED

7 mSvMAX WHOLE-BODY DOSE

250 mGyHAND DOSE (RECOVERY)

5 mSv/hDOSE RATE AT 1 M

4.5 hoursTIME UNDETECTED

£300,000REGULATORY FINE



View RELIR Incident Report +



Transport container & source transfer tools – drill floor setup

INCIDENT TIMELINE

- T + 0:00
Source Transfer Attempted
- T + 5 min
Cable Tension Reads Zero
- T + 30 min **ACTIVE**
AI-CRV Emergency Floor Triggered
Without SiteShield, workers continue routine operations within the undetected exclusion zone. AI-CRV composite score crosses the 85–point emergency floor.
- T + 4.5 hr
Source Discovered — Manual Recovery
- Recovery
Source Returned to Shield

87
AI-CRV

WHAT SITESHIELD PREVENTS

WITHOUT SITESHIELD

- ✗ Source separation: no automated detection
- ✗ Drill floor operations continue normally
- ✗ Zero kN tension: no alert generated
- ✗ 4.5 hours of unmonitored exposure
- ✗ 14 workers irradiated; £300,000 fine

WITH SITESHIELD

- ✓ T+5 min: cable tension = 0 kN fires RADIO alert
- ✓ Source state EXTRACTED triggers base score 50
- ✓ AI-CRV composite reaches 85 — emergency floor
- ✓ RSO auto-notified; drill floor cleared in <5 min
- ✓ Workers never enter unmonitored exclusion zone

TIME TO ALERT COMPARISON



AI-CRV RESPONSE (SIM)

87 / 100

AI-CRV emergency floor — all personnel exit drill floor immediately

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- What is happening right now?
- Should I initiate shutdown?
- What sensors are active?

Ask the safety advise Send

Running...



UAVS DETECTED0

UAVS IN RED ZONE0

AI-CRV SCORE85

MODULES

THREAT --

OPS 42

RIG 99

VEH 55

FLEET 72

RADIO 21

EXPLO 98

EXT THREAT

OPS RISK

RIG FLOOR

VEHICLE

RADIOACTIVE

EXPLOSIVE

SENSORS

Radar

RF Sensor

PTZ Camera

Thermal/IR

SYSTEM

STANDBY

AI-CRV ADVISER

ONLINE

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CLR

Local

Cloud

VICTUS \ llama:7b

Safety adviser ready.
Ask for a situation briefing
or use the quick prompts below.

What is happening right now?

Should I initiate shutdown?

What sensors are active?

Ask the safety advise

Send

Explosive Systems

2 active

EXP-001

Armed

TCP (Tubing Conveyed Perforating) — XN-47H

4 guns x 40 charges = 160 total

Depth: 485 / 500 m MD

EXP-002

Armed

Wireline Conveyed Perforating — BN-09

3 guns x 40 charges = 90 total

Depth: 2890 / 500 m MD

Armed

Detonators live — depth interlock active

Sequence VALID

Safety Interlocks

Depth	Pressure
ACTIVE	ACTIVE
Manual Safety Arm	Firing Auth
LOCKED	STANDBY

Gun Configuration

Gun Count	Total Charges
4	160
Charge Type	Shot Density
Deep Penetrating (DP)	16/m
Manufacturer	
Schlumberger	
Model	
MaxPERF HSD	

Live Parameters

Arming State	Current Depth
ARMED	2890 m MD
Target Depth	Depth Delta
2895 m MD	-5 m
Sequence Integrity	Firing Authorization
VALID	STANDBY

Emergency Procedures

Explosive Alerts

CRITICAL

02:58 PM

UNSAFE ARMING: Armed above safe depth threshold

From: PERF-2026-091

INFO

08:45

TCP gun string running in hole — EXP-001 (160 charges)

From: PERF-2026-091

INFO

08:30

Detonator continuity check PASS — all 160 charges confirmed

From: Firing System

MEDIUM

07:50

Explosives permit validated — license UAE-EXP-2026-447 expires 2026-06-30

From: Regulatory Compliance

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INVESTOR PITCH

Operators Are Drowning in Disconnected Risk Alerts

SCADA alarm

BOP Pressure Low

Radar system

Threat Detected 3km

GPS tracker

Vehicle Speed Violation

Wireline monitor

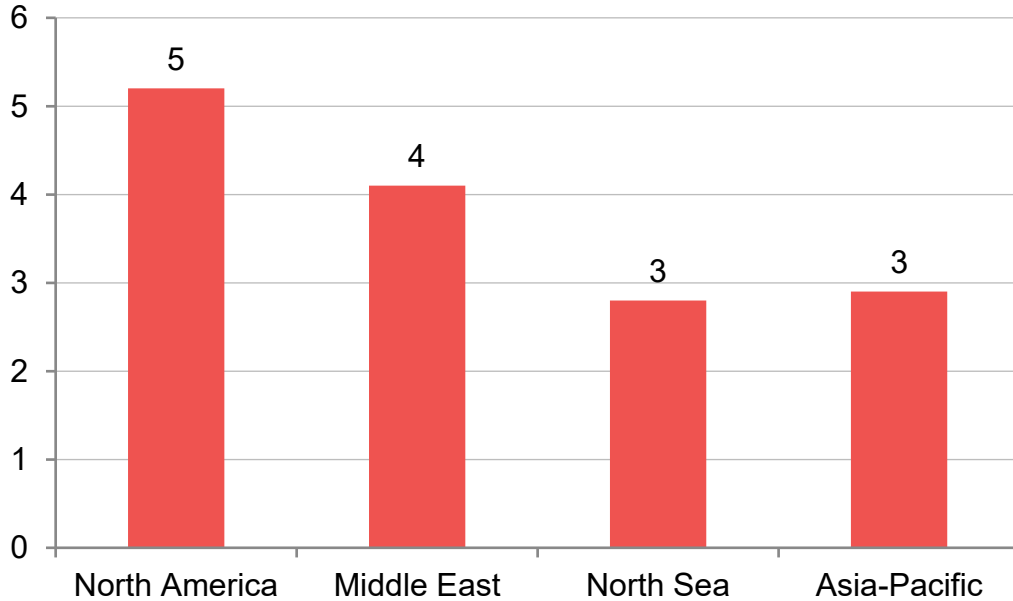
**Source Cable Tension
High**

- Oil & gas operations face **20+ independent** risk domains
- Each domain has separate monitoring systems (SCADA, GPS, CCTV, sensors)
- No unified view = operators miss correlations, false negatives, alert fatigue

"73% of offshore incidents involve MULTIPLE simultaneous risk factors"

— UK HSE Offshore Accident
Report 2023

The Cost of Blind Spots: \$15 Billion in Preventable Losses



\$15B/year

Global NPT in drilling operations

\$500K/day

Average cost per stuck pipe incident

30 min

Average delay between risk emergence and detection

\$10M+

Average cost of a single blowout event

"Current risk management = reactive, not predictive"

SiteShield.ai Modules: Many Combinations, Built for Industrial Workflow

ThreatAware

External Threat Detection

\$20K/year

OpsAware

NPT Prediction

\$12K/year

FleetAware

Vehicle Safety

\$8K/year

HAZMAT Aware

Source Tracking & Safety

\$25K/year

- **2 Modules: 10% discount**
- **3 Modules: 20% discount**
- **Enterprise (all 4): \$45K/year (30% off)**

WHY MODULAR?

Lower entry barrier (\$12K vs. \$45K)
Higher revenue per customer (upsells)
Faster sales cycles (buy one, expand later)

Market Opportunity: \$18B TAM, \$2.1B SAM, \$120M SOM

TAM (Total Addressable Market)

\$18.0B

All drilling & production sites globally (50K sites × \$360K)

SAM (Serviceable Available Market)

\$2.1B

MENA + North America only (6K sites × \$360K)

SOM (Serviceable Obtainable Market)

\$120M

Year 5 target: 2% SAM penetration (333 sites × \$360K)

MARKET DRIVERS:

- NPT reduction mandates from operators (Aramco, ADNOC, Shell)
- HSE regulations tightening (OSHA 1910, ISO 45001)
- Digital transformation budgets growing 15% YoY

Business Model: High Margin, Predictable Revenue

METRIC	VALUE	NOTES
ACV (Average Contract Value)	\$36K	Blended: 1.5 modules/customer
Gross Margin	85%	SaaS: cloud costs only
CAC (Customer Acquisition Cost)	\$8K	Direct sales + trade shows
LTV (Lifetime Value)	\$216K	6-year avg retention
LTV:CAC Ratio	27:1	Best-in-class SaaS metric
Payback Period	3 months	Cash-positive quickly

REVENUE MODEL:

- Annual subscriptions (paid upfront, 12-month contracts)
- Multi-year discounts: 10% (2yr), 20% (3yr) — locks in revenue
- Upsell expansion: 40% of customers add 2nd module within Year 1

Go-To-Market: Land & Expand in MENA

PHASE 1: Beachhead (Year 1)

Direct sales to UAE/Saudi drilling contractors
Trade shows: ADIPEC, SPE ATCE
Pilot-to-paid conversion program

PHASE 2: Scale (Year 2-3)

Partner with SLB/Halliburton for co-marketing
Expand to North America (Permian, Eagle Ford)
Channel sales via regional distributors

PHASE 3: Global (Year 4-5)

Asia-Pacific expansion (Malaysia, Indonesia)
Enterprise deals with IOCs (Shell, BP, Total)
White-label OEM for SCADA vendors

PHASE 4: Diversification (Year 5+)

Adjacent markets: Mining, Renewables
API/data licensing revenue stream
Acquisition exit to strategics

Product Roadmap: From 4 Modules to 10+

Q2 2027

Launch ThreatAware v2.0

AI vision (YOLO) for FOD detection
360° radar integration (DJI AeroScope)

Q3 2027

Launch WeatherAware

Lightning risk prediction
Hurricane/sandstorm alerts

Q4 2027

Launch CyberAware

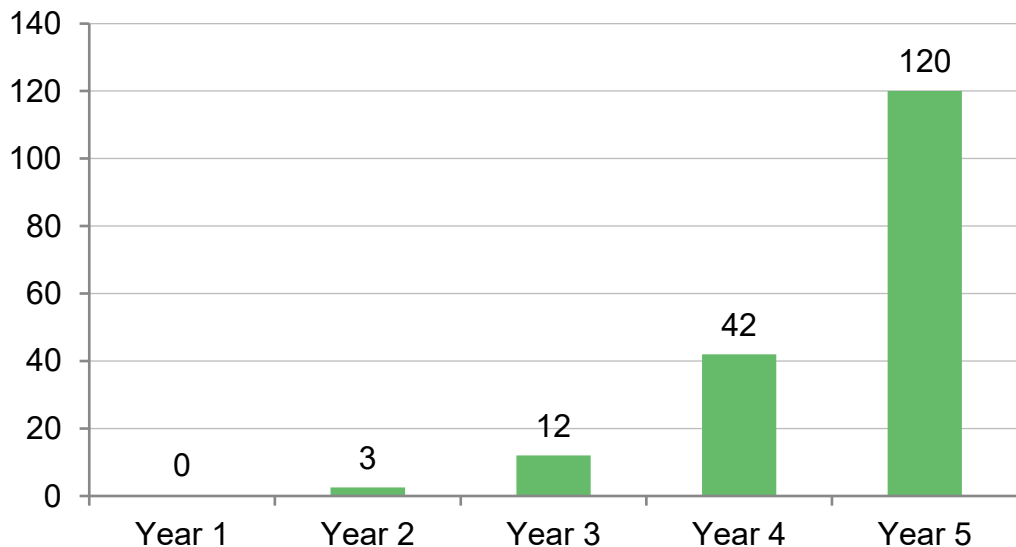
OT/IT network intrusion detection
RANSOMWARE early warning

Q1 2028

API Platform Launch

REST API for 3rd-party integration
Data licensing revenue stream

Financials: \$120M ARR by Year 5

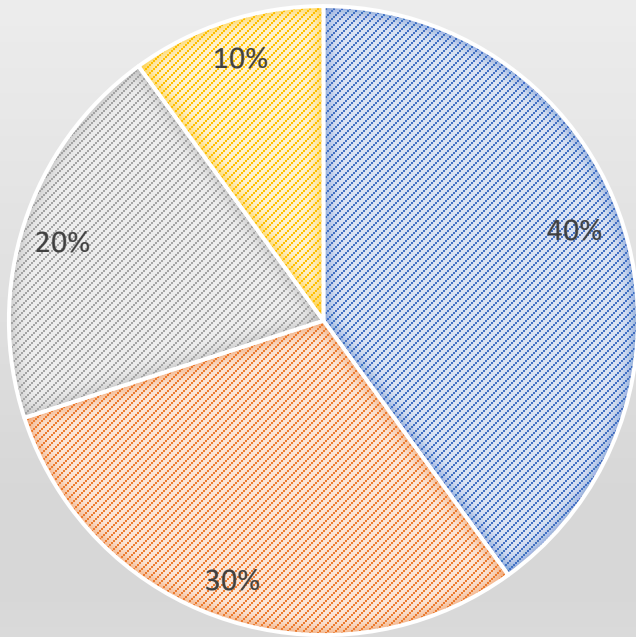


METRIC	YR 1	YR 3	YR 5
Customers	10	80	333
ARR (\$M)	\$0.3	\$12.0	\$120.0
Gross Margin	75%	83%	85%
Burn Rate (\$K/mo)	\$180	\$420	\$850
Headcount	5	25	80

CASH FLOW POSITIVE: Month 48 (Year 4, Q4) — Conservative Model

Use of Funds: \$2M Over 18 Months

- Engineering (40%)
- Sales & Marketing (30%)
- Operations (20%)
- IP & Legal (10%)



DETAILED BREAKDOWN:

Engineering (40%)

- Hire CTO + 4 ML engineers
- Build WeatherAware, CyberAware modules

Sales & Marketing (30%)

- Hire VP Sales + 4 AEs
- ADIPEC booth, digital ads, pilots

Operations (20%)

- AWS/GCP infrastructure, DevOps

IP & Legal (10%)

- PCT filing, national phase entry

Exit Strategy: Strategic Acquisition in 4-5 Years

MOST LIKELY ACQUIRERS:

Schlumberger / SLB	Portfolio gap: no AI risk platform	9/10
Halliburton or Baker	lacks multi-domain fusion	8/10
AIQ (G42 + ADNOC)	AI-first strategy, MENA presence	9/10
Aramco Digital	Building in-house AI capabilities	7/10
Honeywell	SCADA vendor seeking AI upsell	6/10

VALUATION THESIS:

- Year 5 ARR: \$120M
- Exit multiple: 2-3× ARR (SaaS benchmark for \$100M+ ARR companies)
- Exit valuation: \$240M - \$360M

Risks & Mitigation: We've Thought This Through

RISK	MITIGATION
Incumbents (SLB/Halliburton) clone our product	Patent moat (36 months). By then, we have network effects (50+ customers).
Long sales cycles (6-9 months for enterprise)	Start with OpEx modules (\$12K, 60-day cycles). Upsell to enterprise later.
Data access barriers (SCADA vendors lock us out)	OPC-UA + MQTT connectors (industry standard). Operators own their data.
Oil price crash reduces CapEx budgets	Our product REDUCES costs (NPT). We're margin-expansion, not growth CapEx.
Regulatory approval delays (offshore)	Start onshore (no offshore regs). Expand offshore Year 2 after certs.

The Ask: \$2,000,000 to Scale-Up

RAISE SIZE: \$2,000,000

VALUATION: \$20 Mil post-money

EQUITY: 10% (founders retain 90%)

INSTRUMENT: SAFE (Simple Agreement for Future Equity)

USE OF FUNDS: 18-month runway

MILESTONES: 20 customers, 3 new modules, profitability path

*"The industry spends **\$15 Billion** a year on problems we can predict.
Help us turn blind spots into foresight."*